

Haseeb Chattha

70136897

Ass 04

1) $\{w|w\}$ contains at least 8 1's

$S \rightarrow A1A1A1A$

$A \rightarrow 0A|1A|\epsilon$

2) $\{w|w\}$ start and end with same symbol

$S \rightarrow 1A1|0A0|0|1|\epsilon$

$A \rightarrow 0A|1A|\epsilon$

3) $\{w|w\}$ length of w is odd

$S \rightarrow ASA|0|1$

$A \rightarrow 0|1$

4) $\{w/w \text{ length of } w \text{ is odd middle is zero}\}$

$$S \rightarrow 0S0 \mid 1S1$$

$$S \rightarrow 1S0 \mid 0S1$$

$$S \rightarrow 0$$

5) $\{w/w \text{ contains more 1's than 0's}\}$

$$S \rightarrow 1x \mid yS \mid Syx$$

$$X \rightarrow 1x \mid \epsilon$$

$$Y \rightarrow 1y0 \mid 0y1 \mid yx \mid \epsilon$$

6) $\{w/w \text{ is palindrome}\}$

$$S \rightarrow 0S0 \mid 1S1$$

$$S \rightarrow 0 \mid 1 \mid \epsilon$$

7) Empty set

$$S \rightarrow S$$